

4(a)(i)	<i>restriction endonuclease</i> 1 cuts, DNA / plasmid ; <i>DNA ligase</i> 2 joins DNA with the plasmid ; <i>DNA polymerase</i> 3 forms dsDNA or makes a complementary DNA strand ; <i>reverse transcriptase</i> 4 uses mRNA (as a template) to make, cDNA / complementary (ss)DNA ; R converts	4
---------	--	----------

Question	Answer	Marks
4(a)(ii)	any three from 1 for transcription factor(s) to bind ; 2 for RNA polymerase to bind ; 3 (so) gene is, expressed / switched on / activated / transcribed ; 4 to, increase / control, transcription of the, gene of interest / marker gene; 5 to cause transcription with a specific environmental change ; A inducible promoter 6 AVP ; e.g. choose promoter to give transcription in, all tissues / all developmental stages	3
4(b)	plasmid X - no insulin because the gene is, inserted backwards / the wrong way around ; plasmid Y - produces insulin because the gene, is inserted the correct way around / AW ; plasmid Z - no insulin because, the gene was not inserted / the plasmid closed up / it is a non-recombinant plasmid ;	3